

INTERNATIONAL CYBERSECURITY AND DIGITAL FORENSICS ACADEMY

Nasarawa State, Nigeria

LAB REPORT

Lab 05: Building a SCADA Human Machine Interface

Item	Details
Lab Title	Lab 05: Building a SCADA Human Machine Interface
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Course Name	Intro to Critical Infrastructure for IT Learners
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1. Executive Summary

This laboratory moved from watching a SCADA screen to building one. The task was to design an operator screen (HMI) in FUXA for an automated liquid-receiving tank controlled by OpenPLC. The screen had to show the tank level, show whether the fill valve is open or closed, and give the operator a clear picture of what the process is doing.

I created a new FUXA project, set a neutral gray background, and drew the tank, a level bar gauge with a 0 to 10 ft range, a motor-actuated fill valve and the connecting pipe. I started the TankFillSim program in OpenPLC Runtime and connected FUXA to it over Modbus TCP at 192.168.1.10, port 502. I created two tags, Tank Level and Tank Valve Open, and linked them to the bar gauge and the valve graphic. I enabled the connection only after the tags and links were ready.

The completed HMI ran against the live simulation and showed the level rising and falling and the valve graphic changing between closed and open. The design follows high-performance HMI ideas: neutral colours, clear labels, and strong colour kept for abnormal conditions.

2. Lab Objectives

By the end of this lab, I should be able to:

- Explain the role of an HMI and apply P&ID ideas to process pictures.
- Apply high-performance HMI and situational-awareness principles.
- Create a new FUXA project and draw the tank, level indicator, valve and piping.
- Connect FUXA to OpenPLC over Modbus TCP and create the PLC device tags.
- Link the tags to graphics: a 0 to 10 ft level gauge and a valve open/closed animation.
- Operate the finished HMI against the automated tank-fill simulation and confirm it shows the real process.

3. Tools and Resources Used

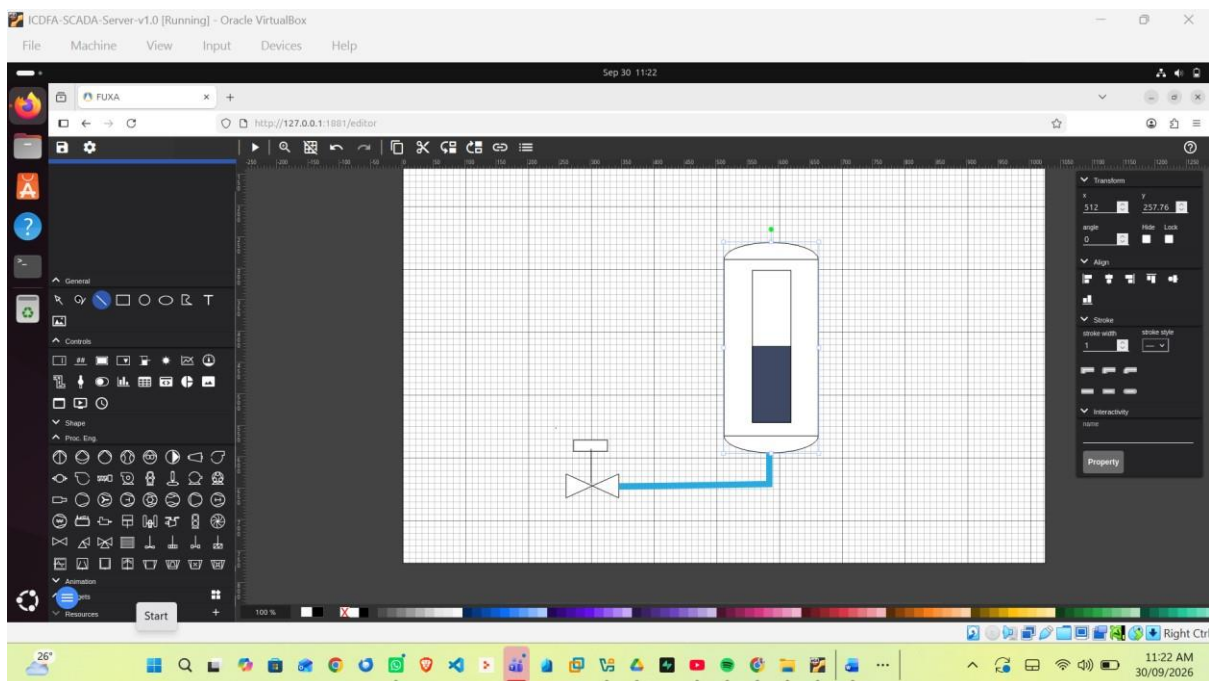
Tool / Resource	What it was used for
PLC VM (192.168.1.10)	Runs OpenPLC Runtime and the tank-fill simulation.
SCADA VM (192.168.2.10)	Runs FUXA, where the HMI was designed and run.
Virtualisation platform: _____ (version _____)	Used to run both VMs on the isolated laboratory network.
FUXA editor (127.0.0.1:1881/editor), version _____	Used to create the project, draw the graphics, set the connection and tags, and run the HMI.
OpenPLC Runtime with TankFillSim.st	The PLC program that fills and drains the tank in simulation.
Modbus TCP, port 502	The industrial protocol used between FUXA and OpenPLC.
Firefox and screenshot tool	Used to open the editor / runtime pages and to save evidence.

4. Methodology (Step-by-Step Procedure)

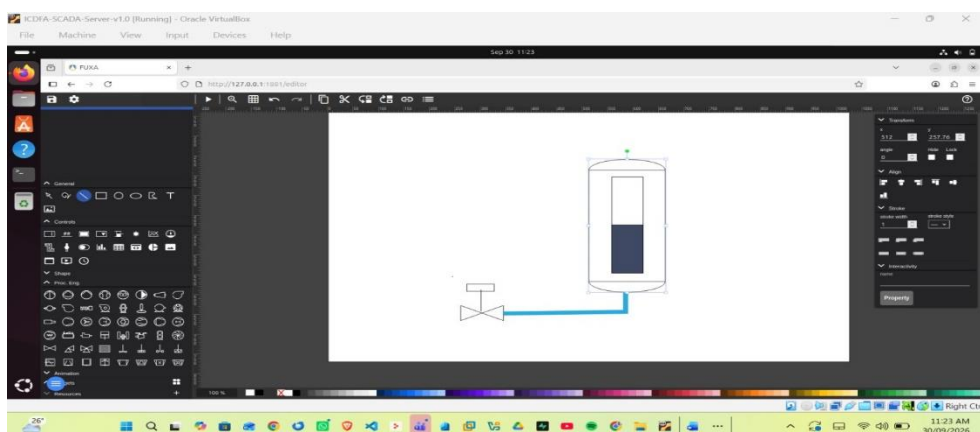
I followed the 14 steps of the lab brief in order. A screenshot was captured at each important step. The screenshots are numbered Figures 1 to 16. Passwords are not shown in this report.

4.2 The Steps

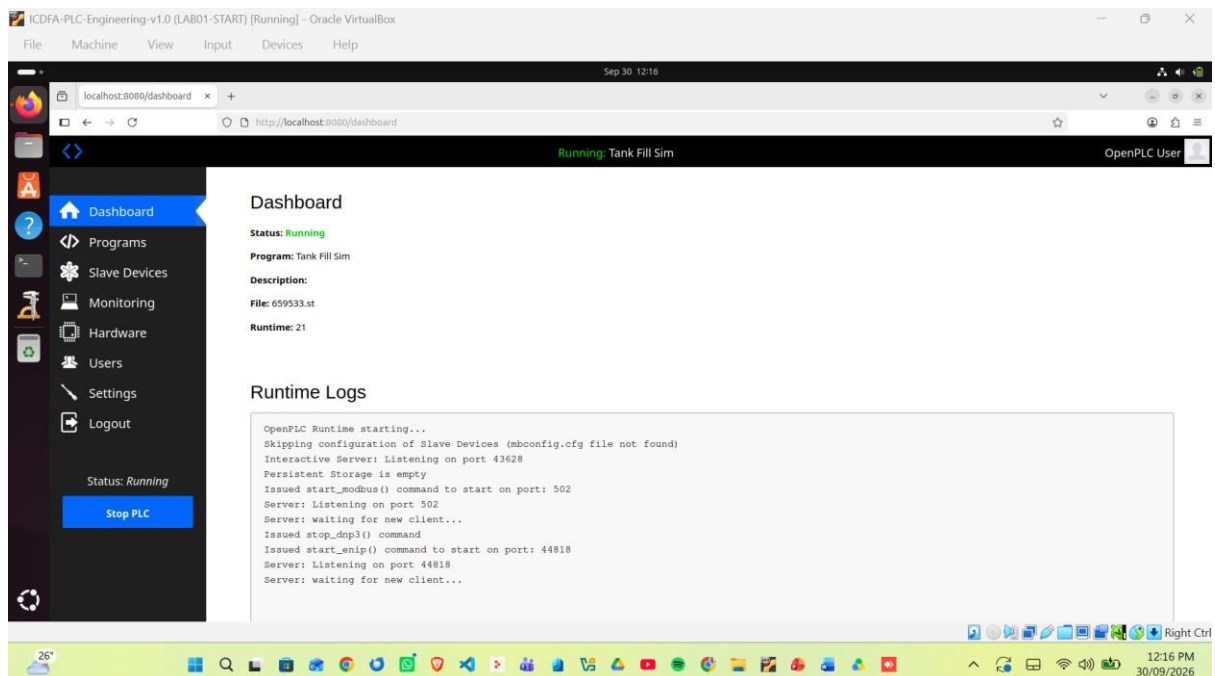
Step 1: Start the VMs and FUXA. I started the PLC VM and SCADA VM on the isolated laboratory network. On the SCADA VM I started FUXA and opened <http://127.0.0.1:1881/editor> in the browser.



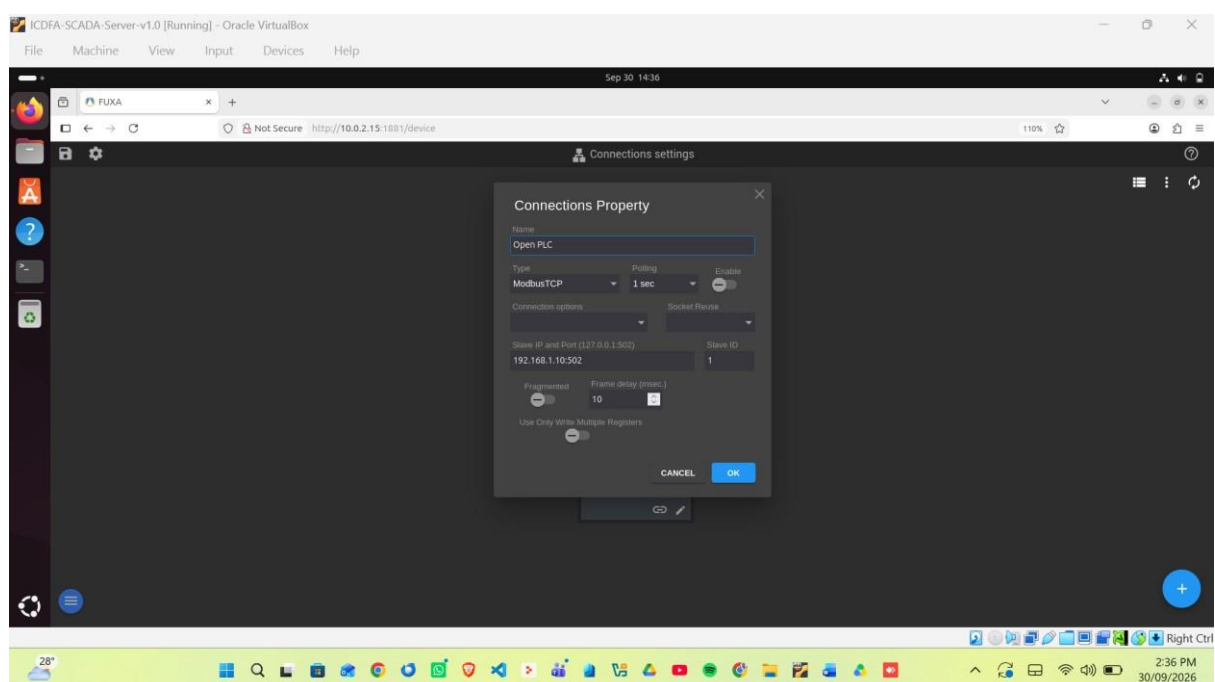
Step 2: Create the project. I created a **New Project** named with my registration number and LAB05: **C11_26_ACIS_17362_LAB05** (slashes are not allowed in names, so I used underscores).



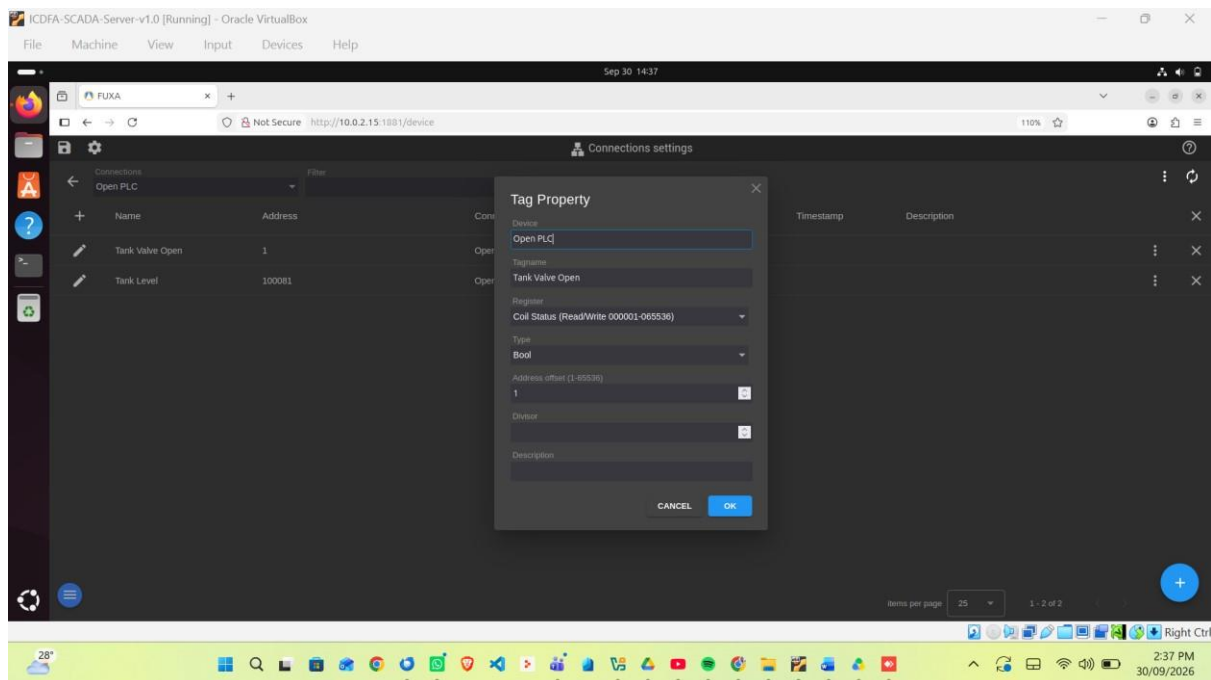
Step 3: Background and grid. I set a **neutral gray** background and turned the **grid** on so that the graphics can be lined up neatly.



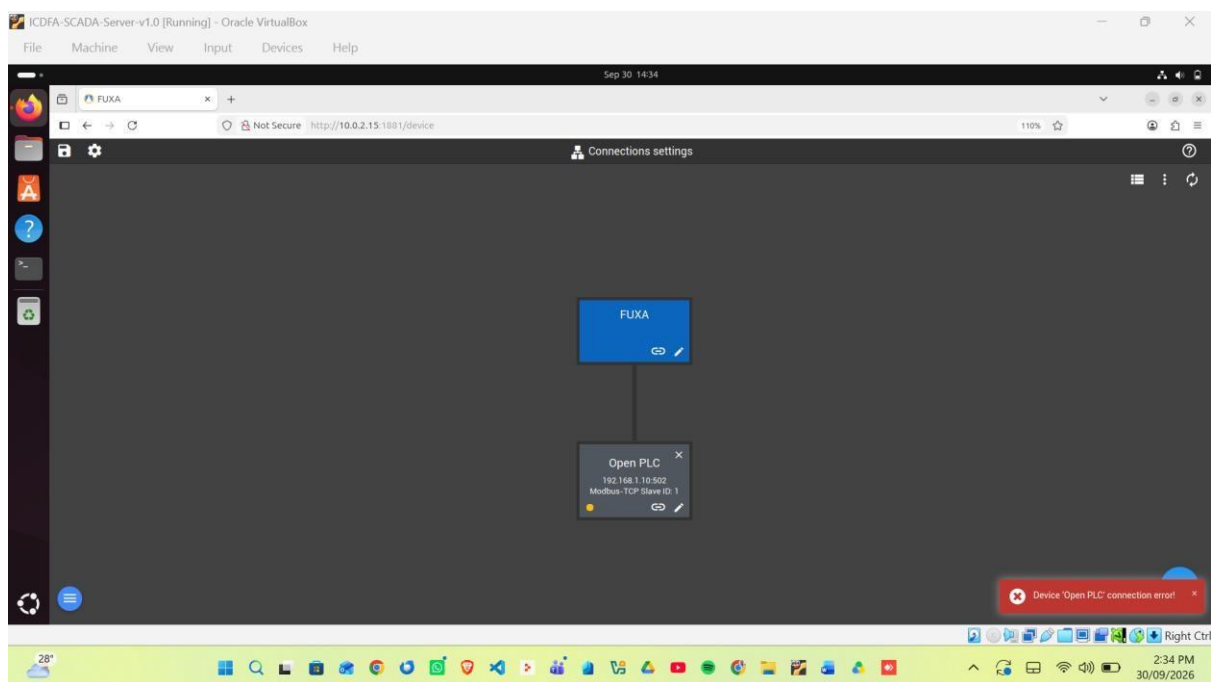
Step 4: Place the tank and the level gauge. From the Process Engineering objects I placed a **tank** graphic. I added a **bar gauge** over / inside the tank and set its engineering range to **0 to 10 ft**.



Step 5: Build the valve and pipe. I built a simple **motor-actuated fill valve** from the available valve / shape objects. I added a **pipe** from the valve to the tank. I kept the normal-state design visually simple.



Step 6: Start the PLC program. On the PLC VM I loaded and started **TankFillSim.st** in OpenPLC Runtime and confirmed that the status is **Running**.



5. Evidence and Analysis

5.1 Evidence 1: Project, Background and Layout

Question: Show the creation of the FUXA project and the configuration of the background and layout (Figures 1 to 3).

Figure	What it shows and why it matters
Figure 1	Both VMs were running on the isolated network, and FUXA was opened in the local editor. This shows the authorised laboratory was used.
Figure 2	A new project was created and named with my registration number and LAB05, so the work is clearly mine and easy to find.
Figure 3	A neutral gray background and the grid were set. Gray is calm and easy on the eyes, and it keeps bright colours free for warnings. The grid helps to line up the objects neatly so the picture is clean and easy to read.

5.2 Evidence 2: Process Graphics (Tank, Level Indicator, Valve and Piping)

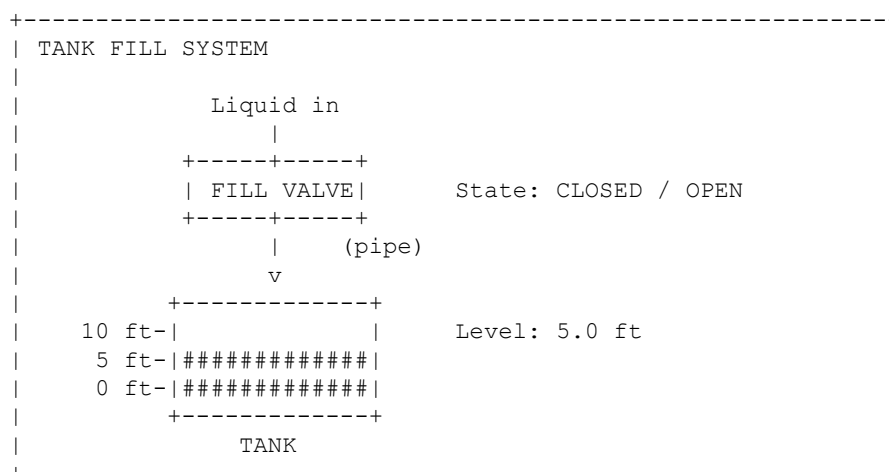
Question: Show the tank, level indicator, motor-actuated valve and piping graphics (Figures 4 and 5).

Part A: The P&ID idea

A **P&ID** (Piping and Instrumentation Diagram) is an engineering drawing that shows the equipment (tanks, pumps), the pipes, the valves and the instruments (such as a level transmitter), and how they are connected. A good HMI follows the same arrangement, so the operator sees the plant in the same order as the drawing: liquid comes in, passes the valve, and enters the tank.

Part B: My layout plan (sketch)

This is my simple sketch before I drew the screen. The real picture is Figure 12.



Part C: What each graphic does

Graphic	Job on the HMI
Tank	Represents the receiving tank. It is the main object the operator watches.

Graphic	Job on the HMI
Bar gauge (0 to 10 ft)	Shows how full the tank is. The scale 0 to 10 ft matches the range of the level transmitter.
Motor-actuated fill valve	Shows the state of the fill valve (open or closed). It is the object that controls the flow into the tank.
Pipe	Joins the valve to the tank and shows the direction of the flow.
Labels	Name each object and show units (ft), so nothing has to be guessed.

5.3 Evidence 3: OpenPLC, Connection and Tags

Question: Show the OpenPLC startup, the FUXA to OpenPLC connection, and the tags (Figures 6 to 9 and 13).

Part A: The PLC is running

Figure 6 shows OpenPLC Runtime with the status **Running**. This means the PLC is executing TankFillSim, which fills and drains the tank. FUXA can only show live values when the PLC is running.

Part B: The connection

Setting	Value
Device name	openplc
Protocol	Modbus TCP
PLC address and port	192.168.1.10 : 502
Role	FUXA is the client that asks; OpenPLC is the server that answers.

Part C: The tags

A **tag** is a name given to one value in the PLC, so the HMI can read it. FUXA reads each tag from the PLC address it is mapped to.

Tag	OpenPLC address	Modbus type and number	Data type	Behaviour
Tank Level	%IW80	Input Register 80	Analogue (a number)	Changes smoothly between 0 and 10 ft. Read only.
Tank Valve Open	%QX0.0	Coil 0	Boolean (0 or 1)	Switches between 0 = closed and 1 = open.

Check on your screen: write the exact wording that your FUXA tag page uses for the type and address, if it is different from this table.

Part D: Why the connection is enabled last

The connection was enabled only after the tags and graphical links were ready (Figure 13). If it is enabled too early, FUXA can start reading half-finished tags and the HMI can show wrong or empty values. The green indicator on the openplc device is the proof that FUXA connected to the PLC successfully.

5.4 Evidence 4: Tag Bindings and Animation

Question: Show the tank-level display with the 0 to 10 range, and the valve open/closed animation (Figures 10 and 11).

Part A: Bar gauge and the 0 to 10 range

The bar gauge is bound to the **Tank Level** tag with **Min = 0** and **Max = 10**. The gauge draws the bar as a fraction of this range.

Worked example: what does the gauge show when the tank level is 7.5 ft?

Step 1: Range = Max - Min = 10 - 0 = 10 ft.

Step 2: Value above Min = 7.5 - 0 = 7.5 ft.

Step 3: Fraction = 7.5 / 10 = 0.75.

Step 4: The bar is $0.75 \times 100 = 75$ percent full.

Answer: At 7.5 ft the bar is three-quarters full. At 2 ft it is 20 percent, and at 10 ft it is full. If the range were set wrongly, for example Max = 100, the bar would show a wrong picture, so the range must match the real 0 to 10 ft scale.

Check on your screen: if your Tank Level value on the HMI is not between 0 and 10, write here the scaling you had to use: _____.

Part B: Valve open / closed animation

The valve graphic is bound to the **Tank Valve Open** tag. The tag has only two values, so the graphic has two looks:

Tag value	Meaning	Valve graphic (my design)
0	Valve CLOSED	Normal look: quiet neutral valve symbol with the label CLOSED.
1	Valve OPEN	Different look that is easy to tell apart (for example filled shape) with the label OPEN.

Check on your screen: describe in the table exactly how your open and closed valves look. The state should be clear from the shape or the label, not from colour alone.

5.5 Evidence 5: The Completed HMI in Operation

Question: Show the completed HMI in design mode and running against the live simulation (Figures 12, 14 and 15).

Part A: What was observed

Write what you actually saw on your screen. Do not copy numbers from anywhere else.

Item observed
Lowest tank level seen (ft)
Highest tank level seen (ft)
Level when the valve graphic changed to OPEN
Level when the valve graphic changed to CLOSED
Does the level rise when the valve shows OPEN? (yes / no)
Does the level fall when the valve shows CLOSED? (yes / no)
Does the HMI value agree with what OpenPLC shows? (yes / no)

Part B: Checking that the HMI shows the real process

Check	What I expect if the HMI is correct
Valve graphic = OPEN	The tank level is rising and the bar gauge grows.
Valve graphic = CLOSED	The tank level is falling as the liquid drains and the bar gauge shrinks.
Level reaches the upper limit	The valve graphic changes to CLOSED.
Level reaches the lower limit	The valve graphic changes to OPEN and filling starts again.

Answer: Figures 14 and 15 show that the level value and the valve graphic change together in real time, so the HMI reflects the live OpenPLC tank-fill simulation. FUXA only reads and shows the values. The PLC still does the control.

5.6 Evidence 6: Exported Project File

Requirement: Submit the exported FUXA project file with the report (Figure 16).

File / Item	Description
This report (PDF)	C11_26_ACIS_17362_LAB05_Building_a_SCADA_Human_Machine_Interface.pdf
Exported FUXA project file	The project C11_26_ACIS_17362_LAB05 exported from FUXA (see Figure 16). Write the exact file name here: _____.

5.7 Evidence 7: Design Rationale (How the HMI Supports Situational Awareness)

The HMI is designed so that an operator can understand the process in a few seconds. The layout follows the flow of the process, in the same way as a P&ID: liquid comes in at the top, passes through the fill valve and pipe, and enters the tank. Because the picture matches the real plant, the operator does not need to think about where things are.

The background is a neutral gray and all normal states are drawn in quiet tones. Strong colour would be kept only for abnormal conditions, so when it appears it catches attention at once. There is no decoration and no unnecessary animation, which reduces distraction and eye strain.

The tank level is shown with a bar gauge against its 0 to 10 ft scale, and the value has its unit. The valve state is shown by its shape and a label, so it does not depend on colour alone. This helps the operator see the present situation, understand it (valve open means the level should be rising), and predict what comes next (the tank is nearly full, so the valve should close soon). Because the values come live from the PLC, the operator always sees the real state of the process.

High-performance HMI ideas used:

Idea	How I applied it
Neutral gray background and quiet normal states	The screen is calm, and nothing competes for attention.
Colour only for abnormal conditions	Strong colour is saved for problems so it stands out.
Show values with units and a scale	The bar gauge and value are read against 0 to 10 ft.
State shown by shape and label	The valve is understood even without colour.
Layout follows the P&ID flow	Liquid in, valve, pipe, tank: the same order as the plant.
No unnecessary effects	Only the level and valve state move.

5.8 Knowledge Check

1. What is a P&ID and how does it influence HMI layout?

A P&ID (Piping and Instrumentation Diagram) is an engineering drawing that shows the equipment, pipes, valves and instruments of a process and how they are connected. It influences the HMI layout because the HMI pictures follow the same arrangement and names, so operators recognise the plant on the screen and can find things quickly.

2. Why do high-performance HMIs avoid unnecessary animation and excessive color?

Too much animation and colour distract the operator, cause tiredness and hide what is important. If everything is bright, a real problem does not stand out. With neutral colours for normal states and colour only for abnormal conditions, the operator notices problems faster and makes fewer mistakes.

3. How does the Tank Level tag differ from the Tank Valve Open tag in data type and behavior?

Tank Level is an analogue value, a number read from Input Register 80 (%IW80). It changes smoothly between 0 and 10 ft and is shown on a scaled bar gauge. Tank Valve Open is a Boolean value, 0 or 1, from Coil 0 (%QX0.0). It jumps between two states, 0 for closed and 1 for open, and is shown as a valve open/closed graphic.

4. What could happen operationally if an HMI displays stale or incorrect PLC data?

The operator may believe the level is normal when the tank is really nearly empty or overflowing, and may take wrong actions or miss a problem. This can cause overflow, spills, damaged equipment, unsafe conditions or an unnecessary shutdown. It also destroys the operator's trust in the screen.

7. Conclusion

In this lab I built my first SCADA HMI. I created a FUXA project, drew a tank, a 0 to 10 ft level gauge, a motor-actuated valve and a pipe, connected FUXA to OpenPLC over Modbus TCP, created the Tank Level and Tank Valve Open tags, and linked them to the graphics. The HMI then showed the live tank-fill simulation, with the level and the valve state changing together. I learned that a good HMI follows the P&ID flow, uses neutral colours and clear labels, and gives the operator the real situation at a glance.

8. Recommendations

- Enable the PLC connection only after the tags and links are ready, and check the green connection indicator.
- Use colour only for abnormal conditions, and always show units and clear labels.
- Show a clear sign on the HMI when communication with the PLC is lost, so the operator is not misled by old data.
- Save and export the project regularly, and keep a clean snapshot of the VMs.
- Keep the laboratory on an isolated network. Modbus TCP has no built-in login or encryption, so it must never be exposed to untrusted networks.

9. References

International Cybersecurity and Digital Forensics Academy (ICDFA). (2026). Lab 05: Building a SCADA Human Machine Interface (Student Edition). Intro to Critical Infrastructure for IT Learners. <https://www.icdfa.edu.ng>

FUXA Project. FUXA web-based process visualization (SCADA/HMI/Dashboard) software. <https://github.com/frangoteam/FUXA>

OpenPLC Project. OpenPLC Runtime documentation. <https://openplcproject.com>

International Society of Automation (ISA). ANSI/ISA-101.01: Human Machine Interfaces for Process Automation Systems.

Hollifield, B., Oliver, D., Nimmo, I., & Habibi, E. (2008). The High Performance HMI Handbook. PAS.